

Text Analysis of Stylometric and Linguistic Patterns in the Bible

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Religious texts have shaped history not only through what they say, but also through how they are written. The four Gospels are often read together because they tell the story of Jesus from four different perspectives. While they overlap in many places, they also differ in structure, emphasis, and wording. These differences make the Gospels a useful case for text analysis, where we can compare not just what each Gospel says, but how each one says it. In this blog post, we will use computational tools to explore whether the Gospels' language patterns reflect their historical and literary relationships.

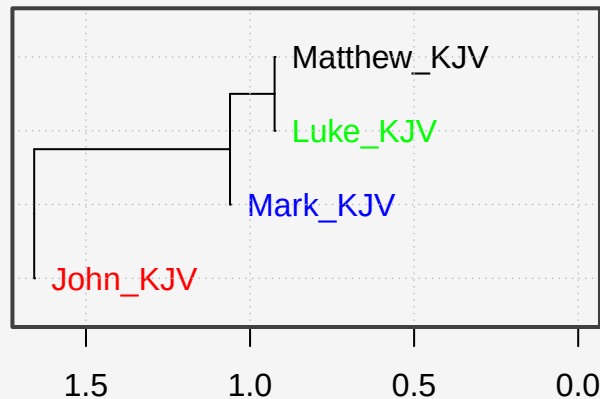
To analyze stylometric (computational and/or quantitative stylistics) patterns in the four Gospels, we used the `stylo` R package. With this package, we wanted to better identify the distinctive writing style among Matthew, Mark, Luke, and John, along with understanding the unique linguistic patterns between them. This is done through `stylos` processing procedures, which measures features such as word frequency and vocabulary. In machine-learning terms, this is an unsupervised analysis, as the algorithm is not trained to predict any known variable, but groups the Gospels based on their stylistic similarities or differences. For this analysis, we used the King James Version texts from Project Gutenberg and compared the four texts specifically using the 300 most frequent words.

Therefore, `stylo` essentially compares how often words are repeated across the texts and then calculates which Gospels are most similar or most different from one another. We should also clarify that, because our analysis uses an English translation, the results should be interpreted as patterns explicitly in the KJV version of the Gospels and may not correlate with other versions.

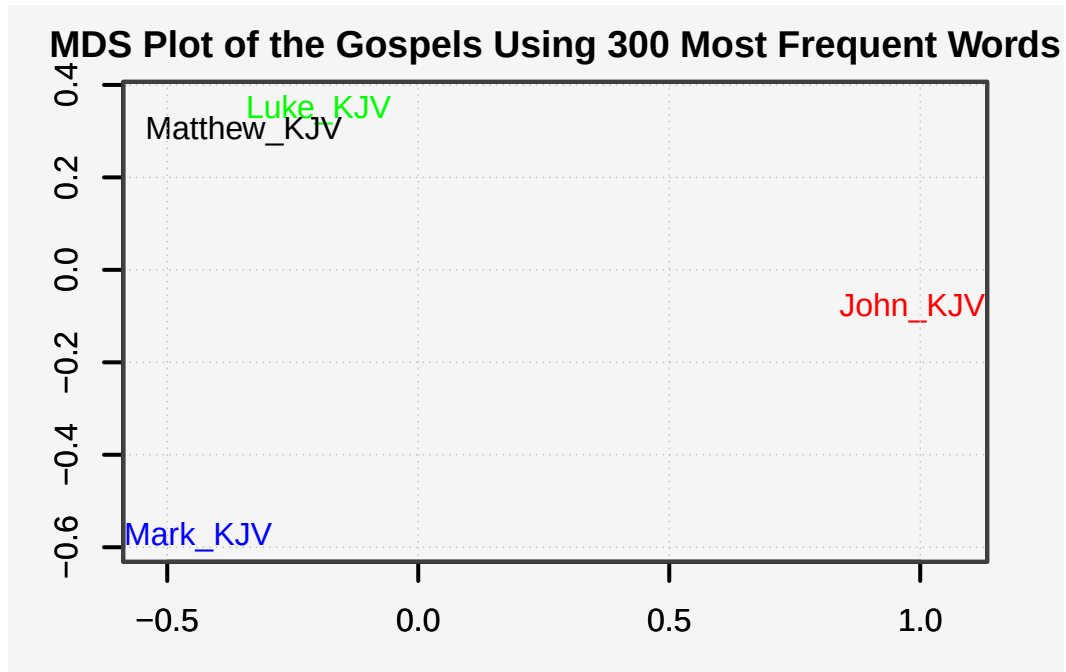
The first plot below is a cluster analysis plot, which works like a family tree of textual similarity. Texts that connect on shorter branches are more similar to one another, while texts that connect farther away are more distant in text patterning. To be more specific, the x-axis shows relative Euclidean distances, which represent stylistic similarity (lower values indicate

greater similarity; higher values indicate greater dissimilarity). Matthew and Luke both sit at roughly 0.8, showing that they are the closest pair in this clustering. John's Gospel seems to be at a much larger distance, demonstrating that it is the most linguistically distinct book in the analysis. This is especially intriguing to me because it aligns with known historical observations about the Gospels. Matthew, Mark, and Luke are known as the Synoptic Gospels, a term derived from the idea of being “seen together”: the “Synoptic Problem” focuses on explaining why they share so many structural parallels. They are grouped this way because they share many of the same stories, structures, and, as we see, even wording patterns. The cluster analysis attempts to quantitatively demonstrate this well-understood belief.

A Plot of the Gospels Using 300 Most Frequent Words



The MDS plot provides a second way to see this same pattern. Unlike the cluster plot, the x- and y-axes in the MDS plot do not represent specific variables. They are instead artificial dimensions created to display stylistic distance. Texts that appear close together have more similar frequent-word patterns, while texts farther apart are more different (similar to the CA). In this MDS plot, Matthew and Luke again appear close together, Mark is somewhat separated but still on the Synoptic side of the comparison, and John sits farthest away. This supports the cluster analysis, which simply shows the same basic relationship in a using the multi-dimensional form.



References/Sources

- https://www.britannica.com/topic/biblical-literature/The-Synoptic-Gospels?utm_source
- web link/document title
- web link/document title
- add more if needed